potash solution fails to reach the top of it and is completely absorbed it shows that only CO_2 is coming and that all air has been expelled from the combustion tube. The nitrometer is then filled with KOH solution by lowering the reservoir and the tap is closed. The combustion tube is now heated in the furnace and the temperature rises gradually. The nitrogen set free form the compound collects in the nitro meter. When the combustion is complete a strong current of CO_2 is sent through, the apparatus in order to sweep the last trace of nitrogen from it. The volume of the gas gets collected is noted after adjusting the reservoir so that the solution in it and the graduated tube is the same. The atmospheric pressure and the temperature are also recorded.

Calculations:

Weight of the substance taken = wg

Volume of nitrogen = V_1 L

Room Temperature = T_1 K

Atmospheric Pressure = P mm of Hg

Aqueous tension at

room temperature = P^1 mm of Hg

Pressure of dry nitrogen = $(P-P^1) = P_1$ mm of Hg.

Let p_0 V_0 and T_0 be the pressure, Volume and temperature respectively of dry nitrogen at STP,

$$\text{Then, } \frac{P_0V_0}{T_0} = \frac{P_1V_1}{T_1}$$

$$\therefore V_0 = \frac{P_1V_1}{T_1} \times \frac{T_0}{P_0}$$

$$V_0 = \left(\frac{P_1V_1}{T_1} \times \frac{273K}{760}\right) \text{mmHg}$$

Calculation of percentage of nitrogen. 22.4 L of N_2 at STP weigh 28gof N_2

$$\therefore V_0 \text{ L of } N_2 \text{ at S.T.P weigh } \frac{28}{22.4} \times V_0$$

Wg of Organic compound contain $\left(\frac{28}{22.4} \times \frac{V_0}{W}\right)$g of nitrogen

∴ Percentage of nitrogen =

$$\left(\frac{28}{22.4} \times \frac{V_0}{W}\right) \times 100$$

Problem: 0.1688 g when analyzed by the Dumas method yield 31.7 mL of moist nitrogen measured at 14^0 C and 758 mm mercury pressure. Determine the % of N in the substance (Aqueous tension at 14^0 C =12 mm)

Weight of Organic compound = 0.168g
Volume of moist nitrogen (V_1) = 31.7mL
= 31.7×10^{-3} L
Temperature (T_1) = 14^0C
= 14 +273
= 287K
Pressure of Moist nitrogen (P) = 758 mm Hg
Aqueous tension at 14^0C = 14^0C
= 12 mm of Hg
∴ Pressure of dry nitrogen = $(P-P^1)$
= 758- 12
= 746 mm of Hg

$$\frac{P_1V_1}{T_1} = \frac{P_0V_0}{T_0}$$

$$\therefore V_0 = \frac{746 \times 31.7 \times 10^{-3}}{287} \times \frac{273}{760}$$

$V_0 = 29.58 \times 10^{-3}$ L

Unit 11 revised - jagan.indd 146 22-01-2021 12:42:30